

Q1:- The temperature 'U' in a metal ball is proportional to the square of the distance from the center of the ball. Find the rate of heat flow across a sphere 'S' of radius 'a' with center at the center of the ball. Use  $\hat{n} = \frac{1}{a}(x\hat{i} + y\hat{j} + z\hat{k})$  as outward unit normal to sphere.

Hint The rate of heat flow across the surface 'S' in the body is then given by the surface integral

$$\iint_S \vec{F} \cdot d\vec{S} = -k \iint_S \nabla U \cdot d\vec{S}$$

Sol Taking the center of the ball to be at the origin, we have

$$U(x, y, z) = C(x^2 + y^2 + z^2)$$

where 'C' is the proportionality constant. Then the heat flow is

$$\vec{F}(x, y, z) = -K \nabla U = -KC(2x\hat{i} + 2y\hat{j} + 2z\hat{k})$$

where 'K' is the conductivity of the metal. The outward unit normal to the sphere  $x^2 + y^2 + z^2 = a^2$  at pt (x, y, z) is

$$\hat{n} = \frac{1}{a}(x\hat{i} + y\hat{j} + z\hat{k})$$

$$\vec{F} \cdot \hat{n} = -\frac{2KC}{a}(x^2 + y^2 + z^2)$$

But on surface we have  $x^2 + y^2 + z^2 = a^2$ , so  $\vec{F} \cdot \hat{n} = -2aKC$ .  
Therefore the rate of heat flow across S is

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} dS$$

$$= -2aKC \iint_S dS$$

$$= -2aKC A(S)$$

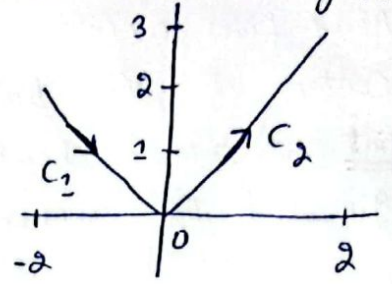
$$= -2aKC (4\pi a^2)$$

$$= -8KC\pi a^3$$

Q2:- Find the work done by the force  $\vec{F}(x,y) = 2xy\hat{i} + 4y^2\hat{j}$  acting along the piecewise smooth curve consisting of line segments from  $(-2, 2)$  to  $(0, 0)$  & from  $(0, 0)$  to  $(2, 3)$ . Also sketch the graph.

sol Work =  $\int_C \vec{F} \cdot d\vec{r}$

$$= \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r}$$



$C_1 : x = t \quad y = -t \quad -2 \leq t \leq 0$

$dx = dt \quad dy = -dt$

$\vec{r}(t) = t\hat{i} - t\hat{j}$

$d\vec{r} = \langle 1, -1 \rangle dt$

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{-2}^0 (-2t^2 - 4t^2) dt$$

$$= \int_{-2}^0 -6t^2 dt$$

$$= -2t^3 \Big|_{-2}^0$$

$$= -2 [0 - (-8)] = -16$$

$C_2 : x = t \quad y = \frac{3}{2}t$

$0 \leq t \leq 2$

$dx = dt \quad dy = \frac{3}{2} dt$

$\vec{r}(t) = t\hat{i} + \frac{3}{2}t\hat{j}$

$d\vec{r} = \langle 1, \frac{3}{2} \rangle dt$



$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 \frac{33}{2} t^2 dt$$

$$= \frac{11}{2} t^3 \Big|_0^2$$

$$= \frac{11}{2} (8-0) = 44$$

$$\text{Work} = \int_C \vec{F} \cdot d\vec{r} = -16 + 44 = 28$$

Q3:- Is it possible for  $f(x,y) = x^2 + x - y$  to be the potential fn of an electrostatic field that is located in a region of  $\mathbb{R}^2$  free of static charge?

Hint The potential fn of an electrostatic field in a region of space that has no static charge is harmonic. For harmonic fn  $\nabla^2 f = 0$ .

Sol If  $f$  were a potential fn, then  $f$  would be harmonic.

$$f_{xx} = 2, \quad f_{yy} = 0$$

$$f_{xx} + f_{yy} \neq 0$$

Therefore,  $f$  is not harmonic &  $f$  cannot represent an electrostatic potential.

Q4:- Calculate the surface integral  $\iint_S \vec{F} \cdot \vec{N} dS$ , where  $\vec{F} = \langle -y, x, 0 \rangle$  &  $S$  is the surface with parametrization  $\vec{r}(u,v) = \langle u, v^2 - u, u+v \rangle$ ,  $0 \leq u \leq 3$ ,  $0 \leq v \leq 4$ .

Sol  $\vec{t}_u = \langle 1, -1, 1 \rangle$        $\vec{t}_v = \langle 0, 2v, 1 \rangle$

$$\vec{t}_u \times \vec{t}_v = \langle -1-2v, -1, 2v \rangle$$

$$\iint_S \vec{F} \cdot d\vec{S} = \int_0^3 \int_0^4 \vec{F}(\vec{r}(u,v)) \cdot (\vec{t}_u \times \vec{t}_v) du dv$$

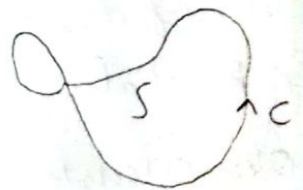
$$\begin{aligned}
&= 4 \int_0^3 \int_0^3 \langle u-v^2, u, 0 \rangle \cdot \langle -1-2v, -1, +2v \rangle du dv \\
&= 4 \int_0^3 \int_0^3 [(u-v^2)(-1-2v) - u] du dv \\
&= 4 \int_0^3 \int_0^3 (2v^3 + v^2 - 2uv - 2u) du dv \\
&= 4 \int_0^3 [2v^3u + v^2u - vu^2 - u^2]_0^3 dv \\
&= 4 \int_0^3 (6v^3 + 3v^2 - 9v - 9) dv \\
&= \left. \frac{3v^4}{2} + v^3 - \frac{9v^2}{2} - 9v \right|_0^4 \\
&= 340
\end{aligned}$$

Flux of  $\vec{F}$  across  $S$  is 340.

Q5:- Consider the 'pringle' surface  $S$  which is part of the graph  $z = xy$  inside the cylinder  $x^2 + y^2 = 2$ . Let  $C$  be the boundary of  $S$ , counterclockwise oriented when viewed from above. Set up, but don't evaluate, a definite integral that computes the line integral  $\int_C z^2 ds$ .

Sol

$$\begin{aligned}
\vec{r}(t) &= \sqrt{2} \cos t \hat{i} + \sqrt{2} \sin t \hat{j} + 2 \sin t \cos t \hat{k} \\
\vec{r}(t) &= \sqrt{2} \cos t \hat{i} + \sqrt{2} \sin t \hat{j} + \sin(2t) \hat{k} \quad 0 \leq t \leq 2\pi \\
\|\vec{r}'(t)\| &= \sqrt{(-\sqrt{2} \sin t)^2 + (\sqrt{2} \cos t)^2 + (2 \cos(2t))^2}
\end{aligned}$$



$$\begin{aligned}
&= \sqrt{2 + 4 \cos^2(2t)} \\
\int_C z^2 ds &= \int_0^{2\pi} \sin^2(2t) \sqrt{2 + 4 \cos^2(2t)} dt
\end{aligned}$$



Q6:- Let  $S$  be the surface of a cylindrical solid  $E$  whose boundary is  $x^2 + y^2 = 4$ ,  $z = 0$  &  $z = 1$ . Let  $\vec{F} = x^3\hat{i} + y^3\hat{j} + z^2\hat{k}$  & let  $\vec{n}$  be the outer unit normal to  $S$ . Use divergence theorem to find  $\iint_S \vec{F} \cdot \vec{n} \, dS$ .

Sol  $\text{div } \vec{F} = 3x^2 + 3y^2 + 2z$

Using Cylindrical Coordinates

$$3x^2 + 3y^2 + 2z = 3r^2 \cos^2 \theta + 3r^2 \sin^2 \theta + 2z$$

$$= 3r^2 + 2z$$

Solid  $E$  is given by  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ ,  $0 \leq z \leq 1$

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \int_0^{2\pi} \int_0^2 \int_0^1 (3r^3 + 2rz) \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 \left[ 3r^3 z + rz^2 \right]_0^1 \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 (3r^3 + r) \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[ \frac{3r^4}{4} + \frac{r^2}{2} \right]_0^2 \, d\theta$$

$$= \int_0^{2\pi} 14 \, d\theta$$

$$= 28\pi$$